

PROPIQ AI Architecture And Code Explanation

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Scope: source code, tests, config, workflows, scripts, and major data/model/BI assets.

Generated/vendor/build/cache folders are excluded. Environment files are listed only by purpose; secret values are not included.

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1. Architecture Overview

PROPIQ AI is a Hyderabad/Telangana real-estate intelligence platform. The frontend is a React/Vite SPA. The backend is FastAPI with SQL persistence, Mongo helpers, CSV/JSON fallback data, ML services, deed verification, PropBot agent APIs, chat, analytics, and Power BI export support.

```
Browser
-> React/Vite SPA: routes, pages, Zustand state, Axios wrappers, Leaflet/Recharts
-> FastAPI /api/v1
    Auth -> SQL User -> JWT
    Properties -> SQL listings + CSV/JSON fallback + static /images
    Predict -> MLService -> pickle models or heuristics
    Deeds -> DeedService -> storage, OCR, fuzzy match, legal checklist
    Agents -> AgentService/LangChain -> Azure OpenAI or fallback replies
    Chat -> WebSocket + SQL ChatMessage
    Analytics -> SQL seller ops + MLService + Power BI-shaped tables
-> Storage/integrations: SQLite/Azure SQL, MongoDB, Redis, Azure OpenAI, Blob Storage,
Document Intelligence, Google Places, ElevenLabs, Power BI
```

Layer Summary

Layer	Role
Frontend pages/components	User workflows and UI for search, map, valuation, deeds, analytics, chat, seller, and admin screens.
Frontend utilities/state	Axios wrappers, mock mode, INR formatting, placeholder image selection, persisted auth, compare state.
API routers	HTTP/WebSocket contract between UI and backend.
Services	Business logic and cloud/local fallback boundaries.
Persistence models	SQLAlchemy tables plus Pydantic document/API schemas.
ML/data engineering	Synthetic training, CSV imports, image scraping, Databricks ETL.
Deployment/BI	Docker, Compose, GitHub Actions, Nginx, Power BI exports.

Main Runtime Flows

- Property browsing: React calls `propertiesApi` ; FastAPI merges SQL listings with fallback catalog data; UI renders cards, details, map pins, price history, and compare state.
- AI valuation: `PredictPage` and detail pages call `/predict` ; `MLService` returns model-backed or heuristic results.

3. Deed verification: `DeedPage` uploads files; `DeedService` stores them, runs OCR/local extraction, fuzzy-matches names, attaches legal checklist text, and exposes admin decisions.
4. PropBot: `AgentChat` calls `/agents/chat`; `AgentService` parses intent and returns replies, navigation cards, and UI commands.
5. Buyer/seller chat: `ChatPage` opens a tokenized WebSocket; backend persists and broadcasts messages.
6. Admin/BI: `AdminPage` loads seller operations, deed summaries, verification rows, and Power BI-shaped tables.

2. Route Maps

Backend API

Endpoint	File	Purpose
<code>/health</code>	<code>backend/app/main.py</code>	Backend health/liveness.
<code>/api/v1/auth/*</code>	<code>backend/app/routers/auth.py</code>	Register, login, refresh.
<code>/api/v1/properties/*</code>	<code>backend/app/routers/properties.py</code>	Listings, map, detail, nearby, price history, moderation.
<code>/api/v1/predict/*</code>	<code>backend/app/routers/predict.py</code>	ML valuation and insight APIs.
<code>/api/v1/deeds/*</code>	<code>backend/app/routers/deeds.py</code>	Document verification, legal utilities, admin review.
<code>/api/v1/agents/*</code>	<code>backend/app/routers/agents.py</code>	PropBot chat, voice, search, document Q&A.
<code>/api/v1/chat/*</code>	<code>backend/app/routers/chat.py</code>	Buyer/seller REST and WebSocket chat.
<code>/api/v1/analytics/*</code>	<code>backend/app/routers/analytics.py</code>	Market/admin analytics and Power BI-shaped data.

Frontend

Route	Component	Purpose
<code>/</code>	<code>LandingPage</code>	Home/search/featured listings.
<code>/properties</code>	<code>PropertiesPage</code>	Filterable listing browser.
<code>/properties/map</code>	<code>MapPage</code>	Leaflet map over GeoJSON.
<code>/properties/:id</code>	<code>PropertyDetailPage</code>	Details, valuation, nearby, EMI, chat CTA.
<code>/properties/:id/chat</code>	<code>ChatPage</code>	Buyer/seller chat.
<code>/list-property</code>	<code>AddPropertyPage</code>	Protected listing wizard.
<code>/predict/commercial</code>	<code>PredictPage</code>	Price/commercial/appreciation tools.
<code>/deeds</code>	<code>DeedPage</code>	Legal workflow tools.
<code>/analytics</code>	<code>AnalyticsPage</code>	Market dashboard.

Route	Component	Purpose
/compare	ComparePage	Compare shortlisted properties.
/login	LoginPage	Auth flow.
/seller	SellerPage	Seller dashboard.
/admin	AdminPage	Admin command center.

3. File Catalog

This report covers 106 text/source/config files and 452 extracted classes, functions, methods, and YAML sections.

AI Agent Layer

File	Lines	Purpose
backend/app/agents/__init__.py	0	Package marker so Python can import this directory as a module.
backend/app/agents/propiq_agents.py	358	LangChain tool-calling agent with tools for property search, valuation, legal Q&A, commercial scoring, and appreciation forecasting.

API Schemas

File	Lines	Purpose
backend/app/schemas/__init__.py	0	Package marker so Python can import this directory as a module.
backend/app/schemas/property.py	100	Source/configuration file used by this layer.

Automated Tests

File	Lines	Purpose
backend/tests/__init__.py	0	Package marker so Python can import this directory as a module.
backend/tests/conftest.py	12	Source/configuration file used by this layer.
backend/tests/test_api.py	189	Main async pytest suite covering backend health, auth, listings, ML, deeds, RERA, analytics, GeoJSON, and agent behavior.

BI Exports

File	Lines	Purpose
powerbi-export/propiq-powerbi-dataset.json	259	JSON configuration or structured data file.

Backend API Routers

File	Lines	Purpose
<code>backend/app/routers/__init__.py</code>	0	Package marker so Python can import this directory as a module.
<code>backend/app/routers/agents.py</code>	150	PropBot API for chat, voice, session history, GUI commands, natural-language search, and legal document Q&A.
<code>backend/app/routers/analytics.py</code>	478	Market and admin analytics API, including Power BI-ready seller operations datasets.
<code>backend/app/routers/auth.py</code>	112	Source/configuration file used by this layer.
<code>backend/app/routers/chat.py</code>	359	Buyer/seller WebSocket chat API with JWT validation, persisted messages, active threads, history, and safety alerts.
<code>backend/app/routers/deeds.py</code>	325	Deed/legal workflow API for uploads, OCR verification, admin review, local file serving, stamp duty, timeline, and RERA checks.
<code>backend/app/routers/predict.py</code>	111	ML prediction API for price, appreciation, commercial score, anomaly detection, and locality insights.
<code>backend/app/routers/properties.py</code>	286	Property API that merges SQL listings with CSV/JSON fallback listings and exposes detail, map, nearby, history, verify, and delete behavior.

Backend App Core

File	Lines	Purpose
<code>backend/app/__init__.py</code>	0	Package marker so Python can import this directory as a module.
<code>backend/app/config.py</code>	137	Typed pydantic-settings configuration for app, DBs, Azure services, JWT, external APIs, and ML paths.
<code>backend/app/database.py</code>	90	Async MongoDB/Motor singleton client, database handle, collection helpers, startup ping, and shutdown cleanup.
<code>backend/app/demo_sellers.py</code>	87	Source/configuration file used by this layer.
<code>backend/app/main.py</code>	101	FastAPI entry point: startup/shutdown lifespan, middleware, static image serving, router registration, and health endpoint.

Backend Data Scripts

File	Lines	Purpose
<code>backend/scripts/__init__.py</code>	0	Package marker so Python can import this directory as a module.
<code>backend/scripts/generate_synthetic_data.py</code>	135	Generates realistic Hyderabad property data and optionally inserts it into SQL.
<code>backend/scripts/import_all_data.py</code>	69	Master data import wizard for property and price-history datasets.

File	Lines	Purpose
backend/scripts/import_price_history.py	106	Imports NHB RESIDEX-style price-history Excel/CSV files into MongoDB.
backend/scripts/import_properties_csv.py	130	Imports supported real-estate CSV schemas into MongoDB.
backend/scripts/scrape_images.py	200	Scrapes property images, validates/resizes them, writes thumbnails, and exports image metadata.
backend/scripts/seed_mock_to_mongo.py	454	Seeds MongoDB with curated sample properties aligned to frontend demo data.

Backend Services

File	Lines	Purpose
backend/app/services/__init__.py	0	Package marker so Python can import this directory as a module.
backend/app/services/agent_service.py	387	Lightweight PropBot service for intent parsing, curated replies, GUI commands, OpenAI calls, sessions, search, and RAG delegation.
backend/app/services/data_service.py	633	CSV/JSON-backed Hyderabad property catalog, normalization, deterministic coordinates/images/sellers, filtering, GeoJSON, and analytics.
backend/app/services/deed_service.py	357	Deed file storage, OCR/local extraction, fuzzy name matching, legal checklist/RAG, and verification serialization.
backend/app/services/geocoding_service.py	66	Nearby POI lookup with Google Places integration and local mock fallback.
backend/app/services/ml_service.py	172	ML inference facade with optional pickle artifacts and heuristic fallbacks for local demos.
backend/app/services/rag_service.py	97	Legal RAG service using FAISS/Azure OpenAI when configured and deterministic legal fallbacks otherwise.

Backend Support

File	Lines	Purpose
backend/.env	secret	Local environment-variable file; values are intentionally redacted.
backend/data/sample_properties.json	35627	JSON configuration or structured data file.
backend/ml/__init__.py	0	Package marker so Python can import this directory as a module.
backend/test_azure_sql.py	60	Source/configuration file used by this layer.
backend/test_integration.py	51	Source/configuration file used by this layer.

Backend Utilities

File	Lines	Purpose
backend/app/utils/__init__.py	0	Package marker so Python can import this directory as a module.
backend/app/utils/logging.py	34	structlog setup with optional Azure Application Insights exporter.
backend/app/utils/security.py	96	Password hashing, JWT generation/decoding, current-user dependency, mock-token fallback, and role guards.

CI/CD

File	Lines	Purpose
.github/workflows/deploy.yml	71	GitHub Actions workflow that tests the backend and builds/pushes the unified Docker image to Azure Container Registry.

Data Assets

File	Lines	Purpose
data/README.md	58	Repository documentation or ignore/configuration file.
data/images/images_manifest.json	602	JSON configuration or structured data file.

Data Engineering

File	Lines	Purpose
data-engineering/databricks-notebooks/01_etl_pipeline.py	101	PySpark/Databricks ETL: raw property JSON to cleaned features and locality price index Delta tables.
data-engineering/scrape_hyderabad.py	262	Hyderabad dataset generator and optional live image scraper from 99acres.
data-engineering/utils/html_helpers.py	44	Robots.txt, polite fetch, BeautifulSoup, and text helper utilities for scraping.

Frontend App Shell

File	Lines	Purpose
frontend/src/App.tsx	62	React route shell with protected routes, navbar, page wiring, and global PropBot widget.
frontend/src/index.css	453	Global design system: color tokens, layout helpers, cards, buttons, badges, forms, navbar, hero, maps, chat widget, tables, skeletons, and responsive rules.
frontend/src/main.tsx	11	React entrypoint that mounts the App and imports global styles.
frontend/src/theme.ts	23	Source/configuration file used by this layer.

File	Lines	Purpose
frontend/src/vite-env.d.ts	2	Source/configuration file used by this layer.

Frontend Components

File	Lines	Purpose
frontend/src/components/Agent/AgentChat.tsx	268	Floating PropBot widget with quick prompts, session chat, voice playback, navigation cards, and GUI command dispatch.
frontend/src/components/Property/PropertyCard.tsx	105	Reusable property listing card with image fallback, price/details badges, navigation, and compare action.
frontend/src/components/UI/Navbar.tsx	84	Role-aware top navigation and auth actions.

Frontend Pages

File	Lines	Purpose
frontend/src/pages/AddPropertyPage.tsx	390	Seller/admin multi-step property listing wizard with deed approval gate and final listing submit.
frontend/src/pages/AdminPage.tsx	845	Admin command center for seller analytics, Power BI exports, listing moderation, and buyer deed verification decisions.
frontend/src/pages/AnalyticsPage.tsx	180	Market analytics dashboard with KPI cards, trends, locality scores, commercial zones, and heatmap chips.
frontend/src/pages/ChatPage.tsx	223	Property chat page using REST history and authenticated WebSocket messages.
frontend/src/pages/ComparePage.tsx	122	Side-by-side compare page for up to three stored property cards.
frontend/src/pages/DeedPage.tsx	337	Legal workflow page for upload, status/timeline, stamp-duty calculation, and RERA check.
frontend/src/pages/LandingPage.tsx	229	Home page with hero search, stats, feature cards, featured listings, top localities, and PropBot CTA.
frontend/src/pages/LoginPage.tsx	95	Login/register form with role selection, auth persistence, and role-based redirects.
frontend/src/pages/MapPage.tsx	144	Leaflet map page over backend GeoJSON with price pins and filters.
frontend/src/pages/PredictPage.tsx	280	Tabbed AI prediction page for price, commercial score, and appreciation forecast.
frontend/src/pages/PropertiesPage.tsx	192	Filterable property listing page with grid/list mode and PropBot filter listener.

File	Lines	Purpose
frontend/src/pages/PropertyDetailPage.tsx	263	Property detail page with gallery, valuation, EMI estimate, nearby places, price chart, and seller chat.
frontend/src/pages/SellerPage.tsx	284	Seller dashboard for owned listings, inquiries, mock views/leads, trends, and listing actions.

Frontend State

File	Lines	Purpose
frontend/src/store/useStore.ts	93	Zustand stores for property list/compare state and persisted auth state.

Frontend Support

File	Lines	Purpose
frontend/.env.local	secret	Local environment-variable file; values are intentionally redacted.
frontend/index.html	20	HTML shell or static HTML file.
frontend/nginx.conf	30	Source/configuration file used by this layer.
frontend/package-lock.json	3827	Generated npm dependency lockfile.
frontend/package.json	35	npm package manifest with frontend scripts and dependencies.
frontend/tsconfig.json	24	TypeScript compiler configuration.
frontend/tsconfig.node.json	2	TypeScript compiler configuration.
frontend/vite.config.ts	23	Source/configuration file used by this layer.

Frontend Utilities

File	Lines	Purpose
frontend/src/utls/api.ts	496	Axios client plus real/mock API wrappers for properties, chat, prediction, agents, analytics, deeds, auth, formatting, and images.
frontend/src/utls/mockData.ts	252	Mock data generators used when VITE_USE MOCK is enabled.

ML Training

File	Lines	Purpose
backend/ml/training/__init__.py	0	Package marker so Python can import this directory as a module.
backend/ml/training/train_all.py	209	Synthetic ML training pipeline for price prediction, commercial scoring, anomaly detection, and appreciation history artifacts.

Mongo/Pydantic Models

File	Lines	Purpose
backend/app/models/__init__.py	0	Package marker so Python can import this directory as a module.
backend/app/models/deed.py	27	Source/configuration file used by this layer.
backend/app/models/property.py	65	Source/configuration file used by this layer.
backend/app/models/user.py	31	Source/configuration file used by this layer.

Repository Config

File	Lines	Purpose
.env	secret	Local environment-variable file; values are intentionally redacted.
.mcp.json	9	JSON configuration or structured data file.
.vscode/settings.json	11	JSON configuration or structured data file.

Repository Root

File	Lines	Purpose
DATA_SOURCES.md	309	Repository documentation or ignore/configuration file.
Dockerfile	50	Multi-stage build: compiles the React frontend, installs the Python backend, serves static files with Nginx, and proxies API traffic to FastAPI.
PROJECT_DOCUMENTATION.md	136	Repository documentation or ignore/configuration file.
README.md	90	Repository documentation or ignore/configuration file.
SELLER_ACCOUNTS_AND_PROPERTIES.md	1030	Repository documentation or ignore/configuration file.
docker-compose.yml	64	Local app, MongoDB, and Redis orchestration with environment wiring for Azure, ML, auth, and external APIs.
docs/generate_architecture_report.py	637	Source/configuration file used by this layer.
test_api.py	27	Source/configuration file used by this layer.

SQL Persistence

File	Lines	Purpose
backend/app/db/__init__.py	4	Package marker so Python can import this directory as a module.
backend/app/db/base.py	7	Source/configuration file used by this layer.
backend/app/db/models.py	135	SQLAlchemy ORM entities for users, properties, leads, chat messages, events, reviews, and deed verifications.

File	Lines	Purpose
backend/app/db/session.py	177	SQLAlchemy engine/session setup, Azure SQL URL normalization, SQLite fallback, table creation, and demo user seeding.

Sample Legal Data Tools

File	Lines	Purpose
backend/data/sample_deeds/README.md	58	Repository documentation or ignore/configuration file.
backend/data/sample_deeds/bulk_upload_deeds.json	129	JSON configuration or structured data file.
backend/data/sample_deeds/create_bulk_upload_files.py	185	Source/configuration file used by this layer.
backend/data/sample_deeds/upload_sample_deeds.py	208	Source/configuration file used by this layer.

4. File-By-File And Function Reference

Each entry includes file purpose and every extracted top-level function/class plus class methods where present.

.env

- Layer: Repository Config
- Lines: not read; secret values redacted
- Purpose: Local environment-variable file; values are intentionally redacted.
- Functions/classes: environment values only; intentionally not included.

.github/workflows/deploy.yml

- Layer: CI/CD
- Lines: 71
- Purpose: GitHub Actions workflow that tests the backend and builds/pushes the unified Docker image to Azure Container Registry.

Line	Symbol	What It Does
1	yaml section name	Function/helper implementing name behavior in this file.
3	yaml section on	Function/helper implementing on behavior in this file.
10	yaml section env	Function/helper implementing env behavior in this file.
16	yaml section jobs	Function/helper implementing jobs behavior in this file.

.mcp.json

- Layer: Repository Config
- Lines: 9
- Purpose: JSON configuration or structured data file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

.vscode/settings.json

- Layer: Repository Config
- Lines: 11
- Purpose: JSON configuration or structured data file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

DATA_SOURCES.md

- Layer: Repository Root
- Lines: 309
- Purpose: Repository documentation or ignore/configuration file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

Dockerfile

- Layer: Repository Root
- Lines: 50
- Purpose: Multi-stage build: compiles the React frontend, installs the Python backend, serves static files with Nginx, and proxies API traffic to FastAPI.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

PROJECT_DOCUMENTATION.md

- Layer: Repository Root
- Lines: 136
- Purpose: Repository documentation or ignore/configuration file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

README.md

- Layer: Repository Root
- Lines: 90
- Purpose: Repository documentation or ignore/configuration file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

SELLER_ACCOUNTS_AND_PROPERTIES.md

- Layer: Repository Root
- Lines: 1030
- Purpose: Repository documentation or ignore/configuration file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/.env

- Layer: Backend Support
- Lines: not read; secret values redacted
- Purpose: Local environment-variable file; values are intentionally redacted.
- Functions/classes: environment values only; intentionally not included.

backend/app/__init__.py

- Layer: Backend App Core
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/agents/__init__.py

- Layer: AI Agent Layer
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/agents/propiq_agents.py

- Layer: AI Agent Layer
- Lines: 358
- Purpose: LangChain tool-calling agent with tools for property search, valuation, legal Q&A, commercial scoring, and appreciation forecasting.

Line	Symbol	What It Does
17	<code>function _get_llm</code>	Private helper for get llm inside this module.
57	<code>function search_properties</code>	Search for properties using natural language.
104	<code>function get_price_prediction</code>	Predict property price using the ML model.
154	<code>function query_legal_docs</code>	Answer questions about land deeds, RERA, stamp duty, legal timelines using the RAG pipeline over legal documents.
187	<code>function get_commercial_score</code>	Score the commercial viability of a land parcel using the ML model.

Line	Symbol	What It Does
238	<code>function get_appreciation_forecast</code>	Forecast property price appreciation over 1, 3, and 5 years.
301	<code>function create_propiq_agent</code>	Builds the LangChain AgentExecutor with PROPIQ tools and prompt.
318	<code>function agent_chat</code>	Chat with the PropBot agent, maintaining session history.

backend/app/config.py

- Layer: Backend App Core
- Lines: 137
- Purpose: Typed pydantic-settings configuration for app, DBs, Azure services, JWT, external APIs, and ML paths.

Line	Symbol	What It Does
17	<code>class Settings</code>	Class grouping data and behavior for settings.
107	<code>method Settings._parse_cors_origins</code>	Method on <code>Settings</code> that handles parse cors origins behavior.
119	<code>method Settings._parse_debug</code>	Method on <code>Settings</code> that handles parse debug behavior.
132	<code>function get_settings</code>	Retrieves or derives get settings for this module.

backend/app/database.py

- Layer: Backend App Core
- Lines: 90
- Purpose: Async MongoDB/Motor singleton client, database handle, collection helpers, startup ping, and shutdown cleanup.

Line	Symbol	What It Does
17	<code>function get_mongo_client</code>	Return the Motor async client (creates it if not yet initialised).
31	<code>function get_database</code>	Return the default PROPIQ database handle.
39	<code>async function connect_db</code>	Called at app startup — verify the connection is alive.
50	<code>async function close_db</code>	Called at app shutdown — gracefully close the Motor client.
61	<code>function get_collection</code>	Shortcut: <code>db[collection_name]</code> .
67	<code>function properties_collection</code>	Function/helper implementing properties collection behavior in this file.
70	<code>function users_collection</code>	Function/helper implementing users collection behavior in this file.
73	<code>function transactions_collection</code>	Function/helper implementing transactions collection behavior in this file.
76	<code>function legal_docs_collection</code>	Function/helper implementing legal docs collection behavior in this file.

Line	Symbol	What It Does
79	function analytics_collection	Function/helper implementing analytics collection behavior in this file.
82	function chat_history_collection	Function/helper implementing chat history collection behavior in this file.
87	async function get_db	FastAPI dependency — inject the database into route handlers.

backend/app/db/__init__.py

- Layer: SQL Persistence
- Lines: 4
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/db/base.py

- Layer: SQL Persistence
- Lines: 7
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
4	class Base	SQLAlchemy ORM model for the base entity/table.

backend/app/db/models.py

- Layer: SQL Persistence
- Lines: 135
- Purpose: SQLAlchemy ORM entities for users, properties, leads, chat messages, events, reviews, and deed verifications.

Line	Symbol	What It Does
14	function _uuid	Private helper for uuid inside this module.
18	class User	SQLAlchemy ORM model for the user entity/table.
31	class Property	SQLAlchemy ORM model for the property entity/table.
59	class Lead	SQLAlchemy ORM model for the lead entity/table.
70	class ChatMessage	SQLAlchemy ORM model for the chat message entity/table.
82	class Event	SQLAlchemy ORM model for the event entity/table.
94	class Review	SQLAlchemy ORM model for the review entity/table.
111	class DeedVerification	SQLAlchemy ORM model for the deed verification entity/table.

backend/app/db/session.py

- Layer: SQL Persistence
- Lines: 177
- Purpose: SQLAlchemy engine/session setup, Azure SQL URL normalization, SQLite fallback, table creation, and demo user seeding.

Line	Symbol	What It Does
21	<code>function _fallback_sqlite_url</code>	Private helper for fallback sqlite url inside this module.
25	<code>function _normalize_database_url</code>	Private helper for normalize database url inside this module.
42	<code>function _resolve_database_url</code>	Private helper for resolve database url inside this module.
51	<code>function _create_engine</code>	Private helper for create engine inside this module.
70	<code>function _create_sqlite_engine</code>	Private helper for create sqlite engine inside this module.
74	<code>function _switch_to_fallback_sqlite</code>	Use local SQLite when the configured SQL database is unavailable in dev.
86	<code>function _ensure_connection</code>	Private helper for ensure connection inside this module.
96	<code>function _seed_demo_users</code>	Private helper for seed demo users inside this module.
144	<code>function init_db</code>	Function/helper implementing init db behavior in this file.
159	<code>function db_session</code>	Function/helper implementing db session behavior in this file.
172	<code>function get_db</code>	Retrieves or derives get db for this module.

backend/app/demo_sellers.py

- Layer: Backend App Core
- Lines: 87
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
11	<code>class DemoSeller</code>	Class grouping data and behavior for demo seller.
82	<code>function seller_for_property_position</code>	Return the seller for a 1-based property position.

backend/app/main.py

- Layer: Backend App Core
- Lines: 101
- Purpose: FastAPI entry point: startup/shutdown lifespan, middleware, static image serving, router registration, and health endpoint.

Line	Symbol	What It Does
22	<code>async function lifespan</code>	Runs startup/shutdown work: preloads property data, initializes SQL tables/demo users, connects to MongoDB if available, and closes Mongo on shutdown.

Line	Symbol	What It Does
99	async function health_check	Returns a small health payload for uptime checks and tests.

backend/app/models/__init__.py

- Layer: Mongo/Pydantic Models
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/models/deed.py

- Layer: Mongo/Pydantic Models
- Lines: 27
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
7	function generate_uuid	Function/helper implementing generate uuid behavior in this file.
10	class DeedVerificationModel	Pydantic model/schema for deed verification model data.

backend/app/models/property.py

- Layer: Mongo/Pydantic Models
- Lines: 65
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
7	function generate_uuid	Function/helper implementing generate uuid behavior in this file.
10	class PropertyModel	Pydantic model/schema for property model data.
43	class LandParcelModel	Pydantic model/schema for land parcel model data.

backend/app/models/user.py

- Layer: Mongo/Pydantic Models
- Lines: 31
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
7	function generate_uuid	Function/helper implementing generate uuid behavior in this file.
10	class UserModel	Pydantic model/schema for user model data.

Line	Symbol	What It Does
23	<code>class UserAlertModel</code>	Pydantic model/schema for user alert model data.

backend/app/routers/__init__.py

- Layer: Backend API Routers
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/routers/agents.py

- Layer: Backend API Routers
- Lines: 150
- Purpose: PropBot API for chat, voice, session history, GUI commands, natural-language search, and legal document Q&A.

Line	Symbol	What It Does
15	<code>class ChatRequest</code>	Pydantic request/response schema used by this router for chat request payloads.
21	<code>class ChatResponse</code>	Pydantic request/response schema used by this router for chat response payloads.
29	<code>class GUICommandRequest</code>	Pydantic request/response schema used by this router for guicommand request payloads.
36	<code>class SearchRequest</code>	Pydantic request/response schema used by this router for search request payloads.
41	<code>class DocQueryRequest</code>	Pydantic request/response schema used by this router for doc query request payloads.
46	<code>class VoiceRequest</code>	Pydantic request/response schema used by this router for voice request payloads.
52	<code>function _tts_text</code>	Private helper for tts text inside this module.
58	<code>async function agent_chat</code>	Send a message to the Universal GUI Agent (PropBot).
73	<code>async function agent_voice</code>	Generate PropBot speech with ElevenLabs while keeping the API key server-side.
119	<code>async function get_session</code>	Retrieve conversation history for an agent session.
128	<code>async function issue_gui_command</code>	Agent issues a GUI command to the frontend (navigate, filter, compare).
137	<code>async function agent_search</code>	Agent-driven natural language property search.
146	<code>async function doc_query</code>	RAG query on legal documents (deeds, RERA, zoning laws).

backend/app/routers/analytics.py

- Layer: Backend API Routers
- Lines: 478
- Purpose: Market and admin analytics API, including Power BI-ready seller operations datasets.

Line	Symbol	What It Does
27	<code>async function market_trends</code>	Price trends by locality over the specified number of months.
37	<code>async function price_heatmap</code>	GeoJSON heatmap for price intensity across Hyderabad.
62	<code>async function top_localities</code>	Top performing localities ranked by metric.
77	<code>async function commercial_zones</code>	Commercial zone intelligence for Hyderabad.
91	<code>function _month_keys</code>	Private helper for month keys inside this module.
104	<code>function _demo_seller_ops</code>	Private helper for demo seller ops inside this module.
254	<code>function _powerbi_embed_url</code>	Private helper for powerbi embed url inside this module.
263	<code>async function admin_seller_ops</code>	Operational admin analytics for sellers, sold deals, leads, and reviews.
446	<code>async function admin_powerbi_dataset</code>	Power BI-ready flat datasets for admin seller operations.

backend/app/routers/auth.py

- Layer: Backend API Routers
- Lines: 112
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
18	<code>class RegisterRequest</code>	Pydantic request/response schema used by this router for register request payloads.
26	<code>class TokenResponse</code>	Pydantic request/response schema used by this router for token response payloads.
35	<code>class RefreshRequest</code>	Pydantic request/response schema used by this router for refresh request payloads.
40	<code>async function register</code>	Register a new user account.
74	<code>async function login</code>	Login with email + password, returns JWT tokens.
102	<code>async function refresh_token</code>	Refresh access token using refresh token (mock).

backend/app/routers/chat.py

- Layer: Backend API Routers
- Lines: 359

- Purpose: Buyer/seller WebSocket chat API with JWT validation, persisted messages, active threads, history, and safety alerts.

Line	Symbol	What It Does
18	<code>class ConnectionManager</code>	Manages WebSocket connections per channel.
21	<code>method ConnectionManager.__init__</code>	Method on <code>ConnectionManager</code> that handles init behavior.
24	<code>async method ConnectionManager.connect</code>	Method on <code>ConnectionManager</code> that handles connect behavior.
28	<code>method ConnectionManager.disconnect</code>	Method on <code>ConnectionManager</code> that handles disconnect behavior.
32	<code>async method ConnectionManager.broadcast</code>	Method on <code>ConnectionManager</code> that handles broadcast behavior.
44	<code>function _channel_property_id</code>	Private helper for channel property id inside this module.
51	<code>function _property_summary</code>	Private helper for property summary inside this module.
71	<code>function _visible_seller_properties</code>	Return properties whose chats this seller/admin may see.
92	<code>function _get_token_from_websocket</code>	Private helper for get token from websocket inside this module.
101	<code>async function _get_user_from_websocket</code>	Private helper for get user from websocket inside this module.
127	<code>async function websocket_chat</code>	Validates token, accepts messages, prevents sender impersonation, persists chat rows, and broadcasts to the channel.
188	<code>async function start_property_chat</code>	Mark a buyer's property chat as active as soon as they open the chat page.
250	<code>async function get_seller_active_chats</code>	Return active chat threads only for properties owned by the seller.
292	<code>async function get_chat_history</code>	Fetch chat history for a channel (JWT protected).
328	<code>async function send_visit_alert</code>	Send 'visit before payment' nudge alert to the buyer (JWT protected).

backend/app/routers/deeds.py

- Layer: Backend API Routers
- Lines: 325
- Purpose: Deed/legal workflow API for uploads, OCR verification, admin review, local file serving, stamp duty, timeline, and RERA checks.

Line	Symbol	What It Does
24	<code>class VerificationDecision</code>	Pydantic request/response schema used by this router for verification decision payloads.
30	<code>function _latest_for_parcel</code>	Private helper for latest for parcel inside this module.

Line	Symbol	What It Does
39	<code>function _can_view_verification</code>	Private helper for can view verification inside this module.
44	<code>async function upload_deed_documents</code>	Receives deed files, creates a verification row, runs OCR/name/legal checks, and returns serialized status.
81	<code>async function get_admin_deed_summary</code>	Admin stats for properties, users, and deed verification automation.
126	<code>async function list_admin_verifications</code>	Admin list of buyer-submitted verification packets and documents.
146	<code>async function update_admin_verification</code>	Function/helper implementing update admin verification behavior in this file.
173	<code>async function get_deed_file</code>	Serve locally stored deed files to the submitter or an admin.
195	<code>async function get_deed_status</code>	Get the current verification status for a parcel.
210	<code>async function trigger_verification</code>	Re-run OCR, name match, and legal checklist for the latest parcel upload.
226	<code>async function get_legal_timeline</code>	Estimate days-to-completion for deed transfer using ML heuristics.
244	<code>async function calculate_stamp_duty</code>	Calculate stamp duty based on state and property value.
269	<code>async function check_rera</code>	Check RERA registration status through the official Telangana RERA source.

backend/app/routers/predict.py

- Layer: Backend API Routers
- Lines: 111
- Purpose: ML prediction API for price, appreciation, commercial score, anomaly detection, and locality insights.

Line	Symbol	What It Does
10	<code>class LandPriceRequest</code>	Pydantic request/response schema used by this router for land price request payloads.
22	<code>class LandPriceResponse</code>	Pydantic request/response schema used by this router for land price response payloads.
30	<code>class AppreciationRequest</code>	Pydantic request/response schema used by this router for appreciation request payloads.
36	<code>class AppreciationResponse</code>	Pydantic request/response schema used by this router for appreciation response payloads.
41	<code>class CommercialScoreRequest</code>	Pydantic request/response schema used by this router for commercial score request payloads.
50	<code>class CommercialScoreResponse</code>	Pydantic request/response schema used by this router for commercial score response payloads.

Line	Symbol	What It Does
57	<code>class AnomalyRequest</code>	Pydantic request/response schema used by this router for anomaly request payloads.
66	<code>class AnomalyResponse</code>	Pydantic request/response schema used by this router for anomaly response payloads.
73	<code>async function predict_land_price</code>	Predict land/property price using XGBoost model.
82	<code>async function predict_appreciation</code>	Forecast price appreciation for 1/3/5 years using Prophet + LSTM.
91	<code>async function predict_commercial_score</code>	Predict commercial viability score (0-100) for a land parcel.
98	<code>async function detect_anomaly</code>	Detect if a listing has suspicious pricing (Isolation Forest).
105	<code>async function get_locality_insights</code>	ML-based locality score: schools, hospitals, transit, safety, growth.

backend/app/routers/properties.py

- Layer: Backend API Routers
- Lines: 286
- Purpose: Property API that merges SQL listings with CSV/JSON fallback listings and exposes detail, map, nearby, history, verify, and delete behavior.

Line	Symbol	What It Does
17	<code>class PropertyCreate</code>	Pydantic request/response schema used by this router for property create payloads.
32	<code>function _property_to_dict</code>	Converts a SQLAlchemy Property row into the frontend JSON shape.
59	<code>async function create_property</code>	Create a new property listing (seller/admin).
108	<code>async function list_properties_endpoint</code>	Applies filters to SQL listings, fetches matching CSV fallback records, deduplicates, and returns a page of results.
196	<code>async function properties_geojson</code>	GeoJSON FeatureCollection of all property pins for map view.
202	<code>async function get_property_endpoint</code>	Get full property detail by ID (DB merged with CSV).
219	<code>async function get_nearby</code>	Nearby POIs for a property — returns NearbyPOI-compatible list.
235	<code>async function get_price_history</code>	Historical price trend for the property's locality.
244	<code>class PropertyPatch</code>	Pydantic request/response schema used by this router for property patch payloads.
249	<code>async function patch_property</code>	Partial update for a property (used by admin approve).
273	<code>async function delete_property</code>	Function/helper implementing delete property behavior in this file.

backend/app/schemas/__init__.py

- Layer: API Schemas
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/schemas/property.py

- Layer: API Schemas
- Lines: 100
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
8	<code>class PropertyBase</code>	Pydantic model/schema for property base data.
34	<code>class PropertyCreate</code>	Pydantic model/schema for property create data.
38	<code>class PropertyUpdate</code>	Pydantic model/schema for property update data.
46	<code>class PropertyResponse</code>	Pydantic model/schema for property response data.
57	<code>class PropertyListResponse</code>	Pydantic model/schema for property list response data.
64	<code>class GeoJSONFeature</code>	Pydantic model/schema for geo jsonfeature data.
70	<code>class GeoJSONCollection</code>	Pydantic model/schema for geo jsoncollection data.
75	<code>class NearbyPOI</code>	Pydantic model/schema for nearby poi data.
83	<code>class PriceHistoryPoint</code>	Pydantic model/schema for price history point data.
90	<code>class ReviewCreate</code>	Pydantic model/schema for review create data.
96	<code>class ReviewResponse</code>	Pydantic model/schema for review response data.

backend/app/services/__init__.py

- Layer: Backend Services
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/services/agent_service.py

- Layer: Backend Services
- Lines: 387
- Purpose: Lightweight PropBot service for intent parsing, curated replies, GUI commands, OpenAI calls, sessions, search, and RAG delegation.

Line	Symbol	What It Does
33	<code>function _build_gui_command</code>	Private helper for build gui command inside this module.
37	<code>function _build_navigation_link</code>	Private helper for build navigation link inside this module.
41	<code>function _format_inr</code>	Private helper for format inr inside this module.
52	<code>function _query_from_params</code>	Private helper for query from params inside this module.
61	<code>function _build_navigation_links</code>	Private helper for build navigation links inside this module.
118	<code>function _compose_curated_reply</code>	Private helper for compose curated reply inside this module.
205	<code>function _parse_intent</code>	Extracts simple user intent and filters from plain English for PropBot.
257	<code>async function _call_azure_openai</code>	Call Azure OpenAI with conversation history.
299	<code>function _fallback_response</code>	Private helper for fallback response inside this module.
327	<code>class AgentService</code>	Class grouping data and behavior for agent service.
328	<code>method AgentService.__init__</code>	Method on <code>AgentService</code> that handles init behavior.
331	<code>async method AgentService.chat</code>	Method on <code>AgentService</code> that handles chat behavior.
356	<code>async method AgentService.get_session</code>	Method on <code>AgentService</code> that handles get session behavior.
362	<code>async method AgentService.issue_gui_command</code>	Method on <code>AgentService</code> that handles issue gui command behavior.
365	<code>async method AgentService.natural_language_search</code>	Method on <code>AgentService</code> that handles natural language search behavior.
373	<code>async method AgentService.rag_query</code>	Method on <code>AgentService</code> that handles rag query behavior.

backend/app/services/data_service.py

- Layer: Backend Services
- Lines: 633
- Purpose: CSV/JSON-backed Hyderabad property catalog, normalization, deterministic coordinates/images/sellers, filtering, GeoJSON, and analytics.

Line	Symbol	What It Does
189	<code>function _stable_hash</code>	Private helper for stable hash inside this module.
193	<code>function _pick_images</code>	Return a deterministic set of relevant image URLs for a property.

Line	Symbol	What It Does
202	<code>function _apply_demo_seller_ownership</code>	Assign one deterministic seller account to each block of 100 demo properties.
212	<code>function _normalize_locality</code>	Private helper for normalize locality inside this module.
222	<code>function _get_coords</code>	Return deterministic coordinates for a property within a Hyderabad locality.
251	<code>function _build_amenities</code>	Private helper for build amenities inside this module.
255	<code>function _row_to_property</code>	Transforms one raw CSV row into the full property object used by the UI.
393	<code>function _load_data</code>	Loads property JSON/CSV once and falls back to synthetic data if loading fails.
441	<code>function _build_fallback</code>	Generate 80 synthetic Hyderabad properties if CSV is unavailable.
490	<code>function get_properties</code>	Retrieves or derives get properties for this module.
540	<code>function get_property_by_id</code>	Retrieves or derives get property by id for this module.
553	<code>function get_localities</code>	Retrieves or derives get localities for this module.
565	<code>function get_geojson</code>	Return all Hyderabad properties as GeoJSON FeatureCollection.
596	<code>function get_analytics</code>	Compute real analytics from the loaded Hyderabad dataset.

backend/app/services/deed_service.py

- Layer: Backend Services
- Lines: 357
- Purpose: Deed file storage, OCR/local extraction, fuzzy name matching, legal checklist/RAG, and verification serialization.

Line	Symbol	What It Does
28	<code>function _use_cloud_integrations</code>	Private helper for use cloud integrations inside this module.
33	<code>function _safe_filename</code>	Private helper for safe filename inside this module.
38	<code>function _json_dump</code>	Private helper for json dump inside this module.
42	<code>function _json_load</code>	Private helper for json load inside this module.
51	<code>async function _upload_to_blob</code>	Stores a deed file in Azure Blob in cloud mode or local storage in demo mode.
96	<code>function _extract_owner_name</code>	Private helper for extract owner name inside this module.
108	<code>async function _run_ocr</code>	Run Azure Document Intelligence or a deterministic local fallback.
165	<code>function _fuzzy_name_match</code>	Compare two names using fuzzy matching.

Line	Symbol	What It Does
176	<code>function serialize_verification</code>	Function/helper implementing serialize verification behavior in this file.
201	<code>class DeedService</code>	Class grouping data and behavior for deed service.
202	<code>async method DeedService.upload_and_create_verification</code>	Upload deed files and create a verification record.
236	<code>async method DeedService.run_verification</code>	Run OCR, fuzzy name matching, and the legal RAG checklist.
275	<code>async method DeedService.run_legal_check</code>	Use the RAG service to attach a legal checklist summary.
319	<code>method DeedService._fallback_legal_summary</code>	Method on <code>DeedService</code> that handles fallback legal summary behavior.
328	<code>async method DeedService.estimate_legal_timeline</code>	Estimate days-to-completion for deed transfer using ML heuristics.

backend/app/services/geocoding_service.py

- Layer: Backend Services
- Lines: 66
- Purpose: Nearby POI lookup with Google Places integration and local mock fallback.

Line	Symbol	What It Does
6	<code>async function get_nearby_pois</code>	Fetch nearby POIs from Google Places API.
46	<code>function _haversine</code>	Private helper for haversine inside this module.
56	<code>function _mock_pois</code>	Private helper for mock pois inside this module.

backend/app/services/ml_service.py

- Layer: Backend Services
- Lines: 172
- Purpose: ML inference facade with optional pickle artifacts and heuristic fallbacks for local demos.

Line	Symbol	What It Does
12	<code>class MLService</code>	Wraps all ML model inference.
19	<code>method MLService.__init__</code>	Method on <code>MLService</code> that handles init behavior.
34	<code>method MLService._load_price_model</code>	Method on <code>MLService</code> that handles load price model behavior.
40	<code>method MLService._load_commercial_model</code>	Method on <code>MLService</code> that handles load commercial model behavior.
46	<code>async method MLService.predict_price</code>	Predict property price.

Line	Symbol	What It Does
73	async method <code>MLService.predict_appreciation</code>	Forecast appreciation using locality growth trends.
94	async method <code>MLService.predict_commercial_score</code>	Compute commercial viability score 0-100.
123	async method <code>MLService.detect_anomaly</code>	Detect anomalous pricing using Isolation Forest heuristic.
140	async method <code>MLService.get_locality_insights</code>	Method on <code>MLService</code> that handles get locality insights behavior.
155	async method <code>MLService.get_price_history</code>	Return 12-month synthetic price history for a locality.

backend/app/services/rag_service.py

- Layer: Backend Services
- Lines: 97
- Purpose: Legal RAG service using FAISS/Azure OpenAI when configured and deterministic legal fallbacks otherwise.

Line	Symbol	What It Does
7	function <code>_cloud_rag_enabled</code>	Private helper for cloud rag enabled inside this module.
12	class <code>RAGService</code>	Class grouping data and behavior for ragservice.
13	method <code>RAGService.__init__</code>	Method on <code>RAGService</code> that handles init behavior.
17	method <code>RAGService._init_rag</code>	Initialize FAISS vector store from legal documents.
69	async method <code>RAGService.query</code>	Method on <code>RAGService</code> that handles query behavior.
81	method <code>RAGService._fallback</code>	Method on <code>RAGService</code> that handles fallback behavior.

backend/app/utils/__init__.py

- Layer: Backend Utilities
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/app/utils/logging.py

- Layer: Backend Utilities
- Lines: 34
- Purpose: structlog setup with optional Azure Application Insights exporter.

Line	Symbol	What It Does
7	function <code>setup_logging</code>	Function/helper implementing setup logging behavior in this file.

backend/app/utils/security.py

- Layer: Backend Utilities
- Lines: 96
- Purpose: Password hashing, JWT generation/decoding, current-user dependency, mock-token fallback, and role guards.

Line	Symbol	What It Does
20	<code>function hash_password</code>	Function/helper implementing hash password behavior in this file.
24	<code>function verify_password</code>	Function/helper implementing verify password behavior in this file.
31	<code>function create_access_token</code>	Creates access token data, UI, or files for this workflow.
41	<code>function create_refresh_token</code>	Creates refresh token data, UI, or files for this workflow.
50	<code>function decode_token</code>	Function/helper implementing decode token behavior in this file.
57	<code>class MockUser</code>	Class grouping data and behavior for mock user.
58	<code>method MockUser.__init__</code>	Method on <code>MockUser</code> that handles init behavior.
63	<code>async function get_current_user</code>	Retrieves or derives get current user for this module.
89	<code>function require_roles</code>	Function/helper implementing require roles behavior in this file.

backend/data/sample_deeds/README.md

- Layer: Sample Legal Data Tools
- Lines: 58
- Purpose: Repository documentation or ignore/configuration file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/data/sample_deeds/bulk_upload_deeds.json

- Layer: Sample Legal Data Tools
- Lines: 129
- Purpose: JSON configuration or structured data file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/data/sample_deeds/create_bulk_upload_files.py

- Layer: Sample Legal Data Tools
- Lines: 185
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
11	<code>function create_bulk_upload_json</code>	Create a JSON file with all sample deeds for bulk upload.

Line	Symbol	What It Does
160	<code>function create_csv_upload_file</code>	Create a CSV file for bulk upload.

backend/data/sample_deeds/upload_sample_deeds.py

- Layer: Sample Legal Data Tools
- Lines: 208
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
14	<code>class DeedUploader</code>	Class grouping data and behavior for deed uploader.
15	<code>method DeedUploader.__init__</code>	Method on <code>DeedUploader</code> that handles init behavior.
20	<code>method DeedUploader.parse_deed_content</code>	Parse deed content and extract structured data.
57	<code>method DeedUploader.create_deed_payload</code>	Create API payload from parsed deed data.
98	<code>method DeedUploader.upload_deed</code>	Upload a single deed to the API.
119	<code>method DeedUploader.upload_all_sample_deeds</code>	Upload all sample deeds from the sample_deeds directory.
167	<code>method DeedUploader.generate_upload_manifest</code>	Generate a JSON manifest of uploaded deeds.
190	<code>function main</code>	Main upload function.

backend/data/sample_properties.json

- Layer: Backend Support
- Lines: 35627
- Purpose: JSON configuration or structured data file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/ml/__init__.py

- Layer: Backend Support
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/ml/training/__init__.py

- Layer: ML Training
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/ml/training/train_all.py

- Layer: ML Training
- Lines: 209
- Purpose: Synthetic ML training pipeline for price prediction, commercial scoring, anomaly detection, and appreciation history artifacts.

Line	Symbol	What It Does
28	<code>function generate_property_dataset</code>	Generate synthetic Hyderabad property dataset for training.
70	<code>function train_price_model</code>	Train XGBoost price prediction model.
112	<code>function train_commercial_model</code>	Train Gradient Boosting commercial viability classifier.
144	<code>function train_anomaly_model</code>	Train Isolation Forest for anomaly detection.
168	<code>function train_appreciation_model</code>	Create locality price index for Prophet (synthetic time series).

backend/scripts/__init__.py

- Layer: Backend Data Scripts
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/scripts/generate_synthetic_data.py

- Layer: Backend Data Scripts
- Lines: 135
- Purpose: Generates realistic Hyderabad property data and optionally inserts it into SQL.

Line	Symbol	What It Does
45	<code>function generate_property</code>	Function/helper implementing generate property behavior in this file.
94	<code>async function main</code>	Function/helper implementing main behavior in this file.

backend/scripts/import_all_data.py

- Layer: Backend Data Scripts
- Lines: 69
- Purpose: Master data import wizard for property and price-history datasets.

Line	Symbol	What It Does
18	<code>function banner</code>	Function/helper implementing banner behavior in this file.
23	<code>async function main</code>	Function/helper implementing main behavior in this file.

backend/scripts/import_price_history.py

- Layer: Backend Data Scripts
- Lines: 106
- Purpose: Imports NHB RESIDEX-style price-history Excel/CSV files into MongoDB.

Line	Symbol	What It Does
17	<code>async function import_price_history</code>	Parses Excel/CSV price history files and imports them into MongoDB.

backend/scripts/import_properties_csv.py

- Layer: Backend Data Scripts
- Lines: 130
- Purpose: Imports supported real-estate CSV schemas into MongoDB.

Line	Symbol	What It Does
19	<code>async function import_all_csvs</code>	Parses all CSVs in properties_dir and imports them into MongoDB.

backend/scripts/scrape_images.py

- Layer: Backend Data Scripts
- Lines: 200
- Purpose: Scrapes property images, validates/resizes them, writes thumbnails, and exports image metadata.

Line	Symbol	What It Does
53	<code>function make_filename</code>	Generate a unique filename from URL hash.
59	<code>function save_image</code>	Download image, save full + thumbnail.
92	<code>function scrape_ddg</code>	Get image URLs using DuckDuckGo search.
106	<code>function scrape_all_images</code>	Scrapes or generates all images data for local datasets.
169	<code>function export_metadata</code>	Create a JSON manifest of all downloaded images.

backend/scripts/seed_mock_to_mongo.py

- Layer: Backend Data Scripts
- Lines: 454
- Purpose: Seeds MongoDB with curated sample properties aligned to frontend demo data.

Line	Symbol	What It Does
439	<code>async function seed_database</code>	Function/helper implementing seed database behavior in this file.

backend/test_azure_sql.py

- Layer: Backend Support
- Lines: 60
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
10	<code>function sqlalchemy_url_to_pyodbc</code>	Function/helper implementing sqlalchemy url to pyodbc behavior in this file.

backend/test_integration.py

- Layer: Backend Support
- Lines: 51
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
5	<code>function _safe_text</code>	Private helper for safe text inside this module.
10	<code>function test_health</code>	Test case that verifies health behavior.
17	<code>function test_properties</code>	Test case that verifies properties behavior.
29	<code>function test_chat</code>	Test case that verifies chat behavior.

backend/tests/__init__.py

- Layer: Automated Tests
- Lines: 0
- Purpose: Package marker so Python can import this directory as a module.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

backend/tests/conftest.py

- Layer: Automated Tests
- Lines: 12
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
7	<code>function event_loop</code>	Create an instance of the default event loop for the test session.

backend/tests/test_api.py

- Layer: Automated Tests
- Lines: 189
- Purpose: Main async pytest suite covering backend health, auth, listings, ML, deeds, RERA, analytics, GeoJSON, and agent behavior.

Line	Symbol	What It Does
16	<code>async function test_health</code>	Test case that verifies health behavior.
24	<code>async function test_demo_admin_login</code>	Test case that verifies demo admin login behavior.
37	<code>async function test_register_allows_same_email_for_buyer_and_seller_but_not_admin</code>	Test case that verifies register allows same email for buyer and seller but not admin behavior.
69	<code>async function test_list_properties</code>	Test case that verifies list properties behavior.
84	<code>async function test_predict_land_price</code>	Test case that verifies predict land price behavior.
96	<code>async function test_commercial_score</code>	Test case that verifies commercial score behavior.
109	<code>async function test_appreciation_forecast</code>	Test case that verifies appreciation forecast behavior.
121	<code>async function test_stamp_duty</code>	Test case that verifies stamp duty behavior.
131	<code>async function test_rera_check_uses_official_manual_workflow_without_fake_registration</code>	Test case that verifies rera check uses official manual workflow without fake registration behavior.
143	<code>async function test_market_trends</code>	Test case that verifies market trends behavior.
150	<code>async function test_geojson</code>	Test case that verifies geojson behavior.
159	<code>async function test_agent_chat_returns_clear_answer_and_navigation_cards</code>	Test case that verifies agent chat returns clear answer and

Line	Symbol	What It Does
		navigation cards behavior.
175	async function test_agent_voice_requires_elevenlabs_key_when_not_configured	Test case that verifies agent voice requires elevenlabs key when not configured behavior.

data-engineering/databricks-notebooks/01_etl_pipeline.py

- Layer: Data Engineering
- Lines: 101
- Purpose: PySpark/Databricks ETL: raw property JSON to cleaned features and locality price index Delta tables.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

data-engineering/scrape_hyderabad.py

- Layer: Data Engineering
- Lines: 262
- Purpose: Hyderabad dataset generator and optional live image scraper from 99acres.

Line	Symbol	What It Does
50	function _stable_coords_seed	Private helper for stable coords seed inside this module.
54	function deterministic_coords	Function/helper implementing deterministic coords behavior in this file.
63	function can_fetch	Function/helper implementing can fetch behavior in this file.
73	function fetch	Function/helper implementing fetch behavior in this file.
80	function normalize_price	Function/helper implementing normalize price behavior in this file.
92	function normalize_area	Function/helper implementing normalize area behavior in this file.
98	function _is_candidate_image_url	Private helper for is candidate image url inside this module.
109	function _extract_image_urls_from_html	Private helper for extract image urls from html inside this module.
128	function _fetch_image_urls_from_listing	Private helper for fetch image urls from listing inside this module.
138	function _scrape_99acres_listing_urls	Private helper for scrape 99acres listing urls inside this module.
158	function _scrape_99acres_image_urls	Private helper for scrape 99acres image urls inside this module.

Line	Symbol	What It Does
185	<code>function scrape_mock</code>	Generate the Hyderabad dataset and optionally scrape live images from 99acres.

data-engineering/utils/html_helpers.py

- Layer: Data Engineering
- Lines: 44
- Purpose: Robots.txt, polite fetch, BeautifulSoup, and text helper utilities for scraping.

Line	Symbol	What It Does
11	<code>function can_fetch</code>	Return True if the site's robots.txt allows the given path.
23	<code>function fetch</code>	GET <code>url</code> with a standard header and a polite pause.
38	<code>function soup_from_response</code>	Convenient wrapper that returns a lxml-parsed soup.
42	<code>function text_or_none</code>	Function/helper implementing text or none behavior in this file.

data/README.md

- Layer: Data Assets
- Lines: 58
- Purpose: Repository documentation or ignore/configuration file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

data/images/images_manifest.json

- Layer: Data Assets
- Lines: 602
- Purpose: JSON configuration or structured data file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

docker-compose.yml

- Layer: Repository Root
- Lines: 64
- Purpose: Local app, MongoDB, and Redis orchestration with environment wiring for Azure, ML, auth, and external APIs.

Line	Symbol	What It Does
1	<code>yaml section services</code>	Function/helper implementing services behavior in this file.
61	<code>yaml section volumes</code>	Function/helper implementing volumes behavior in this file.

docs/generate_architecture_report.py

- Layer: Repository Root
- Lines: 637
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
163	<code>function relpath</code>	Function/helper implementing relpath behavior in this file.
167	<code>function should_include</code>	Function/helper implementing should include behavior in this file.
178	<code>function layer_for</code>	Function/helper implementing layer for behavior in this file.
195	<code>function read_text</code>	Function/helper implementing read text behavior in this file.
208	<code>function clean_sentence</code>	Function/helper implementing clean sentence behavior in this file.
216	<code>function humanize</code>	Function/helper implementing humanize behavior in this file.
222	<code>function fallback_summary</code>	Function/helper implementing fallback summary behavior in this file.
247	<code>function extract_symbols</code>	Function/helper implementing extract symbols behavior in this file.
315	<code>function explain_symbol</code>	Function/helper implementing explain symbol behavior in this file.
359	<code>function build_files</code>	Function/helper implementing build files behavior in this file.
379	<code>function render_markdown</code>	Render helper for the markdown section.
566	<code>function write_outputs</code>	Function/helper implementing write outputs behavior in this file.
626	<code>function main</code>	Function/helper implementing main behavior in this file.

frontend/.env.local

- Layer: Frontend Support
- Lines: not read; secret values redacted
- Purpose: Local environment-variable file; values are intentionally redacted.
- Functions/classes: environment values only; intentionally not included.

frontend/index.html

- Layer: Frontend Support
- Lines: 20
- Purpose: HTML shell or static HTML file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/nginx.conf

- Layer: Frontend Support
- Lines: 30
- Purpose: Source/configuration file used by this layer.

- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/package-lock.json

- Layer: Frontend Support
- Lines: 3827
- Purpose: Generated npm dependency lockfile.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/package.json

- Layer: Frontend Support
- Lines: 35
- Purpose: npm package manifest with frontend scripts and dependencies.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/src/App.tsx

- Layer: Frontend App Shell
- Lines: 62
- Purpose: React route shell with protected routes, navbar, page wiring, and global PropBot widget.

Line	Symbol	What It Does
18	<code>function RequireRole</code>	Route guard that redirects missing or unauthorized roles.
26	<code>function App</code>	React component rendering the app UI.

frontend/src/components/Agent/AgentChat.tsx

- Layer: Frontend Components
- Lines: 268
- Purpose: Floating PropBot widget with quick prompts, session chat, voice playback, navigation cards, and GUI command dispatch.

Line	Symbol	What It Does
5	<code>interface NavLink</code>	TypeScript interface describing the shape of nav link data in this module.
12	<code>interface Message</code>	TypeScript interface describing the shape of message data in this module.
25	<code>function AgentChat</code>	React component rendering the agent chat UI.
56	<code>function stopVoice</code>	Function/helper implementing stop voice behavior in this file.
64	<code>function playVoice</code>	Requests TTS audio for a bot message and plays it in the browser.
116	<code>function send</code>	Sends the user prompt to the agent API, appends bot replies, stores session ID, and dispatches GUI commands.

Line	Symbol	What It Does
146	<code>function renderMessageText</code>	Render helper for the message text section.

frontend/src/components/Property/PropertyCard.tsx

- Layer: Frontend Components
- Lines: 105
- Purpose: Reusable property listing card with image fallback, price/details badges, navigation, and compare action.

Line	Symbol	What It Does
6	<code>interface Props</code>	TypeScript interface describing the shape of props data in this module.
30	<code>function PropertyCard</code>	React component rendering the property card UI.
42	<code>function handleImgError</code>	UI event handler for img error actions.
47	<code>function handleCardClick</code>	UI event handler for card click actions.

frontend/src/components/UI/Navbar.tsx

- Layer: Frontend Components
- Lines: 84
- Purpose: Role-aware top navigation and auth actions.

Line	Symbol	What It Does
4	<code>function Navbar</code>	React component rendering the navbar UI.

frontend/src/index.css

- Layer: Frontend App Shell
- Lines: 453
- Purpose: Global design system: color tokens, layout helpers, cards, buttons, badges, forms, navbar, hero, maps, chat widget, tables, skeletons, and responsive rules.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/src/main.tsx

- Layer: Frontend App Shell
- Lines: 11
- Purpose: React entrypoint that mounts the App and imports global styles.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/src/pages/AddPropertyPage.tsx

- Layer: Frontend Pages
- Lines: 390
- Purpose: Seller/admin multi-step property listing wizard with deed approval gate and final listing submit.

Line	Symbol	What It Does
5	<code>function AddPropertyPage</code>	React component rendering the add property page UI.
36	<code>function handleNext</code>	Advances the listing wizard, enforces deed verification at step 3, and submits the final property payload.

frontend/src/pages/AdminPage.tsx

- Layer: Frontend Pages
- Lines: 845
- Purpose: Admin command center for seller analytics, Power BI exports, listing moderation, and buyer deed verification decisions.

Line	Symbol	What It Does
36	<code>type AdminTab</code>	TypeScript type describing the shape of admin tab data in this module.
38	<code>function getPropertyId</code>	Retrieves or derives get property id for this module.
40	<code>function formatDate</code>	Function/helper implementing format date behavior in this file.
46	<code>function scoreLabel</code>	Function/helper implementing score label behavior in this file.
51	<code>function percent</code>	Function/helper implementing percent behavior in this file.
53	<code>function badgeForStatus</code>	Function/helper implementing badge for status behavior in this file.
61	<code>function downloadFile</code>	Function/helper implementing download file behavior in this file.
71	<code>function toCsv</code>	Function/helper implementing to csv behavior in this file.
78	<code>function escape</code>	Function/helper implementing escape behavior in this file.
87	<code>function AdminPage</code>	React component rendering the admin page UI.
97	<code>function loadDashboard</code>	Loads property, seller analytics, deed stats, verification rows, and Power BI dataset data in parallel.
122	<code>function handleApprove</code>	UI event handler for approve actions.
133	<code>function handleDelete</code>	UI event handler for delete actions.
146	<code>function openDocument</code>	Function/helper implementing open document behavior in this file.
166	<code>function handleVerificationDecision</code>	UI event handler for verification decision actions.
240	<code>function renderSellerAnalytics</code>	Render helper for the seller analytics section.

Line	Symbol	What It Does
425	<code>function renderPowerBI</code>	Render helper for the power bi section.
618	<code>function renderPropertyTable</code>	Render helper for the property table section.
680	<code>function renderDeedTable</code>	Renders buyer verification packets with document-open, approve, and reject actions.

frontend/src/pages/AnalyticsPage.tsx

- Layer: Frontend Pages
- Lines: 180
- Purpose: Market analytics dashboard with KPI cards, trends, locality scores, commercial zones, and heatmap chips.

Line	Symbol	What It Does
7	<code>function AnalyticsPage</code>	React component rendering the analytics page UI.

frontend/src/pages/ChatPage.tsx

- Layer: Frontend Pages
- Lines: 223
- Purpose: Property chat page using REST history and authenticated WebSocket messages.

Line	Symbol	What It Does
6	<code>type ChatMessage</code>	TypeScript type describing the shape of chat message data in this module.
14	<code>function displayTime</code>	Function/helper implementing display time behavior in this file.
19	<code>function ChatPage</code>	React component rendering the chat page UI.
32	<code>function scrollToBottom</code>	Function/helper implementing scroll to bottom behavior in this file.
112	<code>function handleSend</code>	UI event handler for send actions.
135	<code>function labelFor</code>	Function/helper implementing label for behavior in this file.

frontend/src/pages/ComparePage.tsx

- Layer: Frontend Pages
- Lines: 122
- Purpose: Side-by-side compare page for up to three stored property cards.

Line	Symbol	What It Does
20	<code>function ComparePage</code>	React component rendering the compare page UI.
24	<code>function getBest</code>	Retrieves or derives get best for this module.

frontend/src/pages/DeedPage.tsx

- Layer: Frontend Pages
- Lines: 337
- Purpose: Legal workflow page for upload, status/timeline, stamp-duty calculation, and RERA check.

Line	Symbol	What It Does
6	<code>function DeedPage</code>	React component rendering the deed page UI.
22	<code>function checkStatus</code>	Function/helper implementing check status behavior in this file.
32	<code>function calcStamp</code>	Function/helper implementing calc stamp behavior in this file.
37	<code>function checkRera</code>	Function/helper implementing check rera behavior in this file.

frontend/src/pages/LandingPage.tsx

- Layer: Frontend Pages
- Lines: 229
- Purpose: Home page with hero search, stats, feature cards, featured listings, top localities, and PropBot CTA.

Line	Symbol	What It Does
17	<code>function LandingPage</code>	React component rendering the landing page UI.
35	<code>function handleSearch</code>	UI event handler for search actions.

frontend/src/pages/LoginPage.tsx

- Layer: Frontend Pages
- Lines: 95
- Purpose: Login/register form with role selection, auth persistence, and role-based redirects.

Line	Symbol	What It Does
6	<code>function LoginPage</code>	React component rendering the login page UI.
24	<code>function switchMode</code>	Function/helper implementing switch mode behavior in this file.
30	<code>function submit</code>	Function/helper implementing submit behavior in this file.

frontend/src/pages/MapPage.tsx

- Layer: Frontend Pages
- Lines: 144
- Purpose: Leaflet map page over backend GeoJSON with price pins and filters.

Line	Symbol	What It Does
14	<code>function createPriceIcon</code>	Creates price icon data, UI, or files for this workflow.

Line	Symbol	What It Does
20	<code>interface GeoFeature</code>	TypeScript interface describing the shape of geo feature data in this module.
25	<code>function MapController</code>	Function/helper implementing map controller behavior in this file.
31	<code>function MapPage</code>	React component rendering the map page UI.

frontend/src/pages/PredictPage.tsx

- Layer: Frontend Pages
- Lines: 280
- Purpose: Tabbed AI prediction page for price, commercial score, and appreciation forecast.

Line	Symbol	What It Does
7	<code>type Tab</code>	TypeScript type describing the shape of tab data in this module.
9	<code>function PredictPage</code>	React component rendering the predict page UI.
27	<code>function runPricePredict</code>	Runs the price predict workflow and stores the result.
33	<code>function runCommercial</code>	Runs the commercial workflow and stores the result.
39	<code>function runAppreciation</code>	Runs the appreciation workflow and stores the result.
45	<code>function scoreColor</code>	Function/helper implementing score color behavior in this file.

frontend/src/pages/PropertiesPage.tsx

- Layer: Frontend Pages
- Lines: 192
- Purpose: Filterable property listing page with grid/list mode and PropBot filter listener.

Line	Symbol	What It Does
8	<code>function PropertiesPage</code>	React component rendering the properties page UI.
28	<code>function fetchProperties</code>	Builds query params from filters, calls the listing API, and updates loading/list/total state.
57	<code>function handler</code>	UI event handler for r actions.

frontend/src/pages/PropertyDetailPage.tsx

- Layer: Frontend Pages
- Lines: 263
- Purpose: Property detail page with gallery, valuation, EMI estimate, nearby places, price chart, and seller chat.

Line	Symbol	What It Does
6	<code>function PropertyDetailPage</code>	React component rendering the property detail page UI.

frontend/src/pages/SellerPage.tsx

- Layer: Frontend Pages
- Lines: 284
- Purpose: Seller dashboard for owned listings, inquiries, mock views/leads, trends, and listing actions.

Line	Symbol	What It Does
26	<code>function SellerPage</code>	React component rendering the seller page UI.
41	<code>function fetchDashboardListings</code>	Function/helper implementing fetch dashboard listings behavior in this file.
58	<code>function handleEdit</code>	UI event handler for edit actions.
63	<code>function handleDelete</code>	UI event handler for delete actions.

frontend/src/store/useStore.ts

- Layer: Frontend State
- Lines: 93
- Purpose: Zustand stores for property list/compare state and persisted auth state.

Line	Symbol	What It Does
2	<code>interface Property</code>	TypeScript interface describing the shape of property data in this module.
20	<code>interface Filters</code>	TypeScript interface describing the shape of filters data in this module.
29	<code>interface PropertyStore</code>	TypeScript interface describing the shape of property store data in this module.
37	<code>object method setProperties</code>	Function/helper implementing set properties behavior in this file.
38	<code>object method setLoading</code>	Function/helper implementing set loading behavior in this file.
39	<code>object method setFilters</code>	Function/helper implementing set filters behavior in this file.
40	<code>object method setPage</code>	Function/helper implementing set page behavior in this file.
41	<code>object method addToCompare</code>	Function/helper implementing add to compare behavior in this file.
42	<code>object method removeFromCompare</code>	Function/helper implementing remove from compare behavior in this file.
43	<code>object method clearCompare</code>	Function/helper implementing clear compare behavior in this file.
53	<code>object method setProperties</code>	Function/helper implementing set properties behavior in this file.
55	<code>object method setLoading</code>	Function/helper implementing set loading behavior in this file.

Line	Symbol	What It Does
56	object method setFilters	Function/helper implementing set filters behavior in this file.
57	object method setPage	Function/helper implementing set page behavior in this file.
58	object method addToCompare	Function/helper implementing add to compare behavior in this file.
66	object method removeFromCompare	Function/helper implementing remove from compare behavior in this file.
67	object method clearCompare	Function/helper implementing clear compare behavior in this file.
69	interface AuthStore	TypeScript interface describing the shape of auth store data in this module.
73	object method setAuth	Function/helper implementing set auth behavior in this file.
74	object method logout	Function/helper implementing logout behavior in this file.
79	object method user	Function/helper implementing user behavior in this file.
82	object method setAuth	Function/helper implementing set auth behavior in this file.
87	object method logout	Function/helper implementing logout behavior in this file.

frontend/src/theme.ts

- Layer: Frontend App Shell
- Lines: 23
- Purpose: Source/configuration file used by this layer.

Line	Symbol	What It Does
21	type ColorKey	TypeScript type describing the shape of color key data in this module.

frontend/src/utils/api.ts

- Layer: Frontend Utilities
- Lines: 496
- Purpose: Axios client plus real/mock API wrappers for properties, chat, prediction, agents, analytics, deeds, auth, formatting, and images.

Line	Symbol	What It Does
14	function delay	Function/helper implementing delay behavior in this file.
15	function mock	Function/helper implementing mock behavior in this file.
46	object method list	API wrapper that lists/searches records for this service group.
63	object method get	API wrapper that fetches one record by ID.
68	object method geojson	API wrapper that fetches map-ready GeoJSON.
73	object method nearby	API wrapper that fetches nearby places.

Line	Symbol	What It Does
81	object method priceHistory	API wrapper that fetches price-history data.
89	object method create	API wrapper that creates a new record.
94	object method update	API wrapper that updates a record.
99	object method patch	API wrapper that partially updates a record.
104	object method delete	API wrapper that deletes a record.
113	object method startPropertyChat	Function/helper implementing start property chat behavior in this file.
117	object method history	Function/helper implementing history behavior in this file.
122	object method sellerActive	Function/helper implementing seller active behavior in this file.
131	object method landPrice	Function/helper implementing land price behavior in this file.
135	object method appreciation	Function/helper implementing appreciation behavior in this file.
140	object method commercialScore	Function/helper implementing commercial score behavior in this file.
145	object method anomaly	Function/helper implementing anomaly behavior in this file.
150	object method localityInsights	Function/helper implementing locality insights behavior in this file.
159	object method chat	API wrapper for PropBot chat.
166	object method search	Function/helper implementing search behavior in this file.
177	object method docQuery	Function/helper implementing doc query behavior in this file.
182	object method voice	API wrapper for PropBot text-to-speech audio.
193	object method marketTrends	Function/helper implementing market trends behavior in this file.
197	object method heatmap	Function/helper implementing heatmap behavior in this file.
202	object method topLocalities	Function/helper implementing top localities behavior in this file.
207	object method commercialZones	Function/helper implementing commercial zones behavior in this file.
212	object method demandForecast	Function/helper implementing demand forecast behavior in this file.
221	object method adminSellerOps	Function/helper implementing admin seller ops behavior in this file.
268	object method powerBIDataset	Function/helper implementing power bidataset behavior in this file.
323	object method upload	API wrapper for deed document upload.
347	object method status	API wrapper for deed status.
352	object method verify	API wrapper to trigger deed verification.
357	object method timeline	API wrapper for legal timeline.
362	object method stampDuty	API wrapper for stamp-duty calculation.
367	object method rera	API wrapper for RERA lookup.
372	object method adminSummary	Function/helper implementing admin summary behavior in this file.

Line	Symbol	What It Does
387	object method adminVerifications	Function/helper implementing admin verifications behavior in this file.
411	object method updateVerification	Function/helper implementing update verification behavior in this file.
416	object method documentBlob	Function/helper implementing document blob behavior in this file.
424	type AuthPayload	TypeScript type describing the shape of auth payload data in this module.
429	function mockAuthResponse	Function/helper implementing mock auth response behavior in this file.
444	object method login	API wrapper for role-aware login.
451	object method register	API wrapper for buyer/seller registration.
465	function formatINR	Function/helper implementing format inr behavior in this file.

frontend/src/utis/mockData.ts

- Layer: Frontend Utilities
- Lines: 252
- Purpose: Mock data generators used when VITE_USE MOCK is enabled.

Line	Symbol	What It Does
35	function MOCK_PROPERTY_DETAIL	Function/helper implementing mock property detail behavior in this file.
71	function genTrend	Function/helper implementing gen trend behavior in this file.
127	function MOCK_PRICE_PREDICTION	Function/helper implementing mock price prediction behavior in this file.
137	function MOCK_COMMERCIAL_SCORE	Function/helper implementing mock commercial score behavior in this file.
144	function MOCK_APPRECIATION	Function/helper implementing mock appreciation behavior in this file.
175	function MOCK_AGENT_CHAT	Function/helper implementing mock agent chat behavior in this file.
203	function MOCK_DEED_STATUS	Function/helper implementing mock deed status behavior in this file.
212	function MOCK_DEED_TIMELINE	Function/helper implementing mock deed timeline behavior in this file.
220	function MOCK_STAMP_DUTY	Function/helper implementing mock stamp duty behavior in this file.
238	function MOCK_RERA	Function/helper implementing mock rera behavior in this file.

frontend/src/vite-env.d.ts

- Layer: Frontend App Shell
- Lines: 2
- Purpose: Source/configuration file used by this layer.

- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/tsconfig.json

- Layer: Frontend Support
- Lines: 24
- Purpose: TypeScript compiler configuration.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/tsconfig.node.json

- Layer: Frontend Support
- Lines: 2
- Purpose: TypeScript compiler configuration.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

frontend/vite.config.ts

- Layer: Frontend Support
- Lines: 23
- Purpose: Source/configuration file used by this layer.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

powerbi-export/propiq-powerbi-dataset.json

- Layer: BI Exports
- Lines: 259
- Purpose: JSON configuration or structured data file.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

test_api.py

- Layer: Repository Root
- Lines: 27
- Purpose: Source/configuration file used by this layer.
- Functions/classes: none extracted. This is configuration, styles, static documentation, package metadata, package marker content, or structured data.

5. Data, Model, And BI Assets

Asset	Role
<code>backend/ml/models/*.pkl</code>	Serialized trained/fallback ML artifacts for price prediction, commercial scoring, anomaly detection, and price history.
<code>backend/ml/models/training_data.csv</code>	Synthetic training data emitted by the ML training script.
<code>backend/data/sample_properties.json</code>	Large generated property catalog used by backend fallback/demo loading.
<code>data/datasets/properties/*.csv</code>	Raw/generated property datasets, including Hyderabad and multi-city sources.
<code>data/datasets/price_history/nhb_residex_quarterly.xlsx</code>	NHB RESIDEX quarterly price index source.
<code>data/legal_docs/**/*.*pdf</code>	RERA, building rules, title, registration, stamp, and transfer law documents for legal context/RAG.
<code>data/legal_docs/deeds/telangana_sale_deed_template.txt</code>	Template/legal text used by deed workflows.
<code>data/images/property_photos/*</code>	Property photos served through backend static image mount or referenced in data.
<code>data/images/thumbnails/*</code>	Generated thumbnails for property photos.
<code>data/images/images_manifest.json</code>	Image metadata manifest.
<code>powerbi-export/*</code>	Power BI CSV, dataset JSON, PBIDS, and workbook exports for admin analytics.

6. Deployment And Environment Notes

- Local development can run backend/frontend separately or use Docker Compose.
- The production-style Dockerfile builds React first, then serves frontend static files and FastAPI through a single Nginx/Python image.
- GitHub Actions currently runs backend tests and builds/pushes the Docker image to Azure Container Registry on `main`.
- Environment variables control SQL/Mongo/Redis, Azure OpenAI, Blob Storage, Document Intelligence, Cognitive Search, Google Places, ElevenLabs, JWT, and Application Insights.
- Secret files are intentionally redacted in this report. Keep actual keys in environment variables or Azure Key Vault.

7. Testing Summary

- `backend/tests/test_api.py` is the main automated suite using `httpx.AsyncClient` with ASGI transport.

- It covers health, auth, registration role rules, properties, ML endpoints, stamp duty, RERA manual verification behavior, analytics, GeoJSON, and PropBot behavior.
- `backend/test_integration.py` , `backend/test_azure_sql.py` , and root `test_api.py` are manual smoke/diagnostic scripts against running services.

8. Fast Reading Order

1. `frontend/src/App.tsx` for user routes.
2. `frontend/src/utils/api.ts` for frontend-to-backend mapping.
3. Matching files in `backend/app/routers/` for API contracts.
4. `backend/app/services/` for business logic and fallback behavior.
5. `backend/app/db/models.py` , `backend/app/models/*` , and `backend/app/schemas/*` for data shapes.
6. `backend/tests/test_api.py` for executable examples of expected behavior.